Which of the following sets of quantum number is not possible ?

a) (i), (ii), (iii) and (iv) b) (ii), (iv) and (v)

c) (i) and (iii) d) (ii), (iii) and (iv)

18. How many electrons in an atom with atomic number 105 can have (n + l) = 8 ?

a) 30 b) 17 c) 15 d) unpredictable

19. Electron density in the yz plane of $3d_{xy}$ orbital is

a) zero b) 0.50 c) 0.75 d) 0.90

20. If uncertainty in position and momentum are equal, then minimum uncertainty in velocity is

a) $\frac{1}{m}\sqrt{\frac{h}{\pi}}$ d) $\sqrt{\frac{h}{\pi}}$ c) $\frac{1}{2m}\sqrt{\frac{h}{\pi}}$ d) $\frac{h}{4\pi}$

21. A macroscopic particle of mass 100 g and moving at a velocity of 100 cm s^{-1} will have a de Broglie wavelength of

a) 6.6×10^{-29} cm b) 6.6×10^{-30} cm c) 6.6×10^{-31} cm d) 6.6×10^{-32} cm

22. The ratio of de Broglie wavelengths of a deuterium atom to that of an α - particle, when the velocity of the former is five times greater than that of later, is

a) 4 b) 0.2 c) 2.5 d) 0.4

23. The energy of an electron in the 3rd orbit of hydrogen atom is –E. The energy of an electron in the first orbit will be

a) –3E b) $-E/3$ c) $-E/9$ d) –9E

24. Time independent Schnodinger wave equation is

a) $\hat{H}\psi = E\psi$ b) $\nabla^2\psi + \frac{8\pi^2 m}{h^2}(E+V)\psi = 0$

c) $\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2} + \frac{2m}{h^2}(E-V)\psi = 0$ d) all of these

25. Which of the following does not represent the mathematical expression for the Heisenberg uncertainty principle ?

a) $\Delta x \,.\, \Delta p \geq h/4\pi$ b) $\Delta x \,.\, \Delta v \geq h/4\pi m$

c) $\Delta E \,.\, \Delta t \geq h/4\pi$ d) $\Delta E \,.\, \Delta x \geq h/4\pi$

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